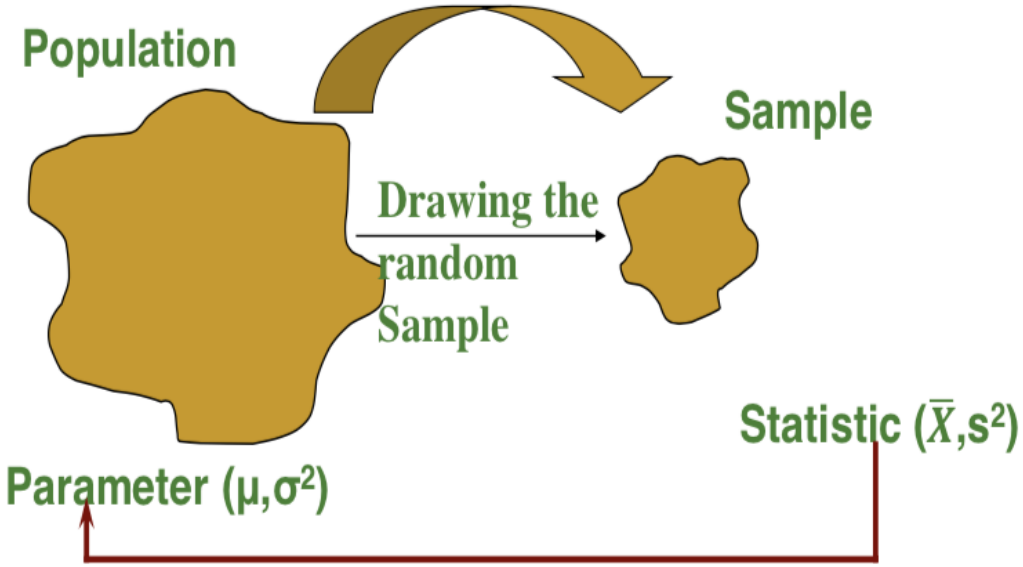
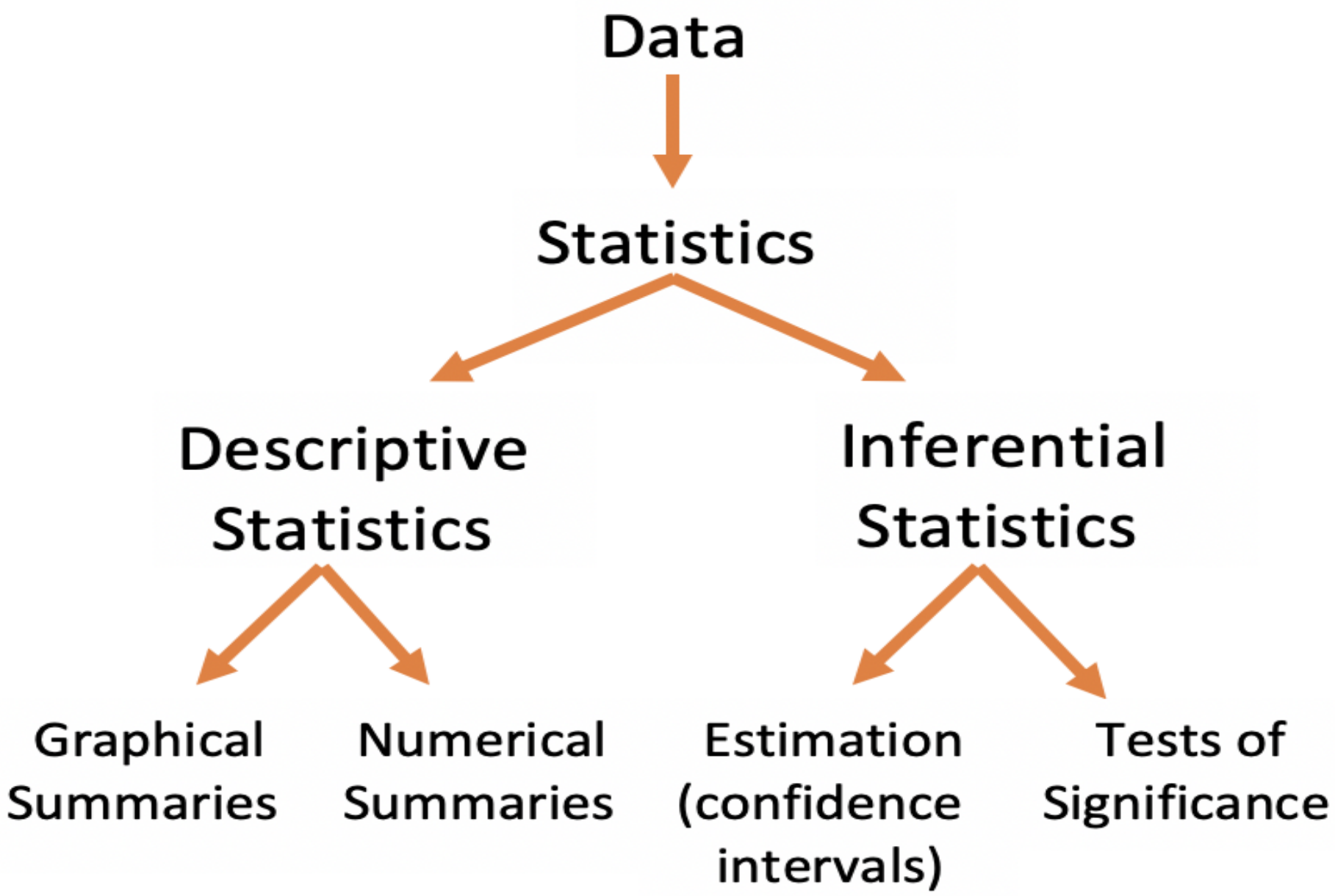
Unit 3: Probability Distributions and Data Simulation

Recall

- Recall that, in the area of statistics, there is some population that we are interested in; we take samples to learn about that population.
- Based on the sample (data) we calculate statistics which tell us about the population characteristics, which are called parameters, i.e.



Furthermore...



A review of probability

- Descriptive statistics allow us to present data in a summarized form.
- We can use various graphical displays and numerical measures to investigate:
  - measures of location
  - measure of variability and shape
  - quantities such as quartiles or percentiles etc.
- The bridge between descriptive statistics and inferential statistics is probability theory, which is the mathematical tool that allows us to link the two parts of statistics.

Randomness and probability

- Statistics involves the study of variability. But how can we work with something that involves so much uncertainty?

- To understand this, we consider the idea of random behaviour.

  “Random" in statistics does not mean something “haphazard” or “chaotic”, but a description of a kind of order that emerges in the long-run.

- In other words, behaviour is unpredictable in the short-run, but has a regular and predictable distribution in the long-run.

Motivating examples

- Suppose you toss a coin a very large number of times. Eventually, the proportion of times you get heads would approach 0.5, and the same for tails (assuming you have a fair coin).
- Consider rolling a die. You don't know what will come up when you roll it but nevertheless a regular pattern would emerge after many repeated rolls.
- We handle such uncertainty by grasping and understanding what are called randomness and probability.

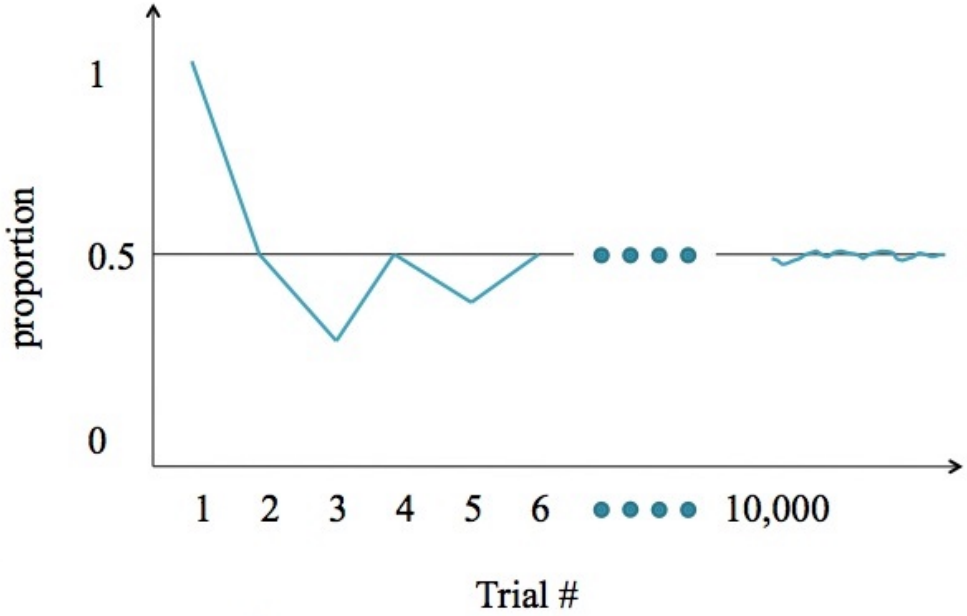
Randomness

- We call a phenomenon random if individual outcomes are uncertain but there is nonetheless a regular distribution of outcomes in a large number of repetitions.

- Let us look at tossing a coin again. Suppose we repeatedly toss a coin and observe the following sequence: H T T H T H. We can record the proportion of Heads after each toss: 1.0, 0.50, 0.33, 0.50, 0.40, 0.50. In time, if you kept tossing, a pattern will emerge.

Example cont'd

- Suppose you toss a coin 10,000 times. In doing so, you will almost surely observe between 4,900 and 5,100 Heads, i.e., the proportion will approach 0.5.
- Visually we can observe this proportion:



- Observations such as this form the foundation for the concept of probability.

Probability

- The probability of any outcome of a random phenomenon is the proportion of times the outcome would occur in a very long series of repetitions.

- What is the difference between proportions and probability?

- A proportion is a known or observed value, while a probability is a theoretical value of a proportion after an infinitely long series of trials. We speak of proportions in the present tense, whereas probability relates to future events.

Defining a probability model

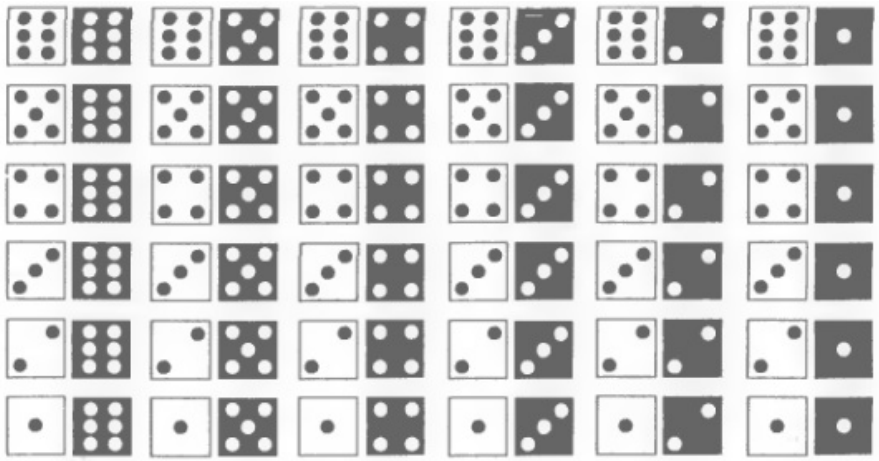
- Since probability cannot be observed, i.e., we cannot toss a coin forever, we must deal with mathematical models; the mathematical model used to describe random behaviour is called a *probability model*.

- A description of a random phenomenon in the language of mathematics is called a probability model.

- A probability model has two components:
  1. list of all possible outcomes, called the sample space, S;
  2. a probability for each outcome.

Example 1

- Suppose we roll two dice and observe the number of dots facing up on each of them.
- The sample space has 36 outcomes: S={(6,6), (6,5),. . . , (6,1), (5,6), . . . , (5,1), . . . , (1,6), . . . ,(1,1)}, where the first number in each outcome represents the number of dots on the top face of the first die (white) and the second number represents the number of dots on the top face of the second die (black). Observe $36 = 6 \times 6$.



(Gonick and Smith, 1993)

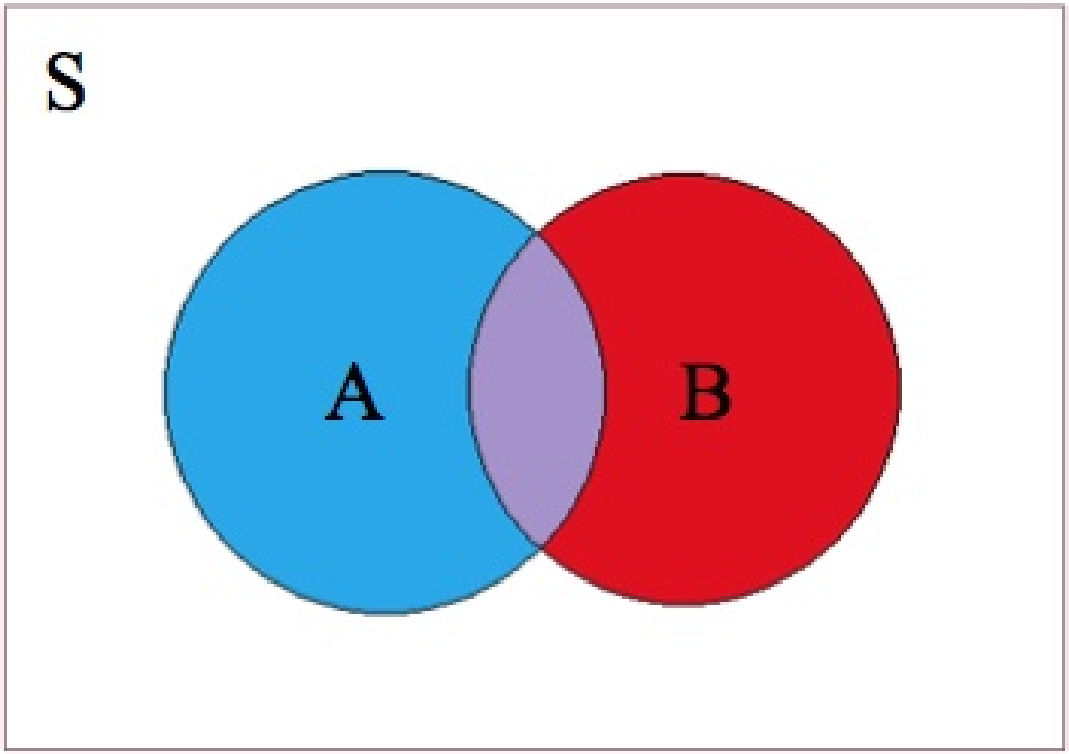
Note: We only make them different colours to tell them apart.

Events

- An event is an outcome or a set of outcomes of a random phenomenon. That is, an event is a *subset* of the sample space S.
- Consider a coin being tossed three times and observing Head (H) or a Tail (T). Suppose we are interested in the *event* A where exactly two Tails are observed. What outcomes are in A?
- In this case, A={TTH, HTT, THT} and A is an event.

Union of events

- The union of two events A and B, denoted by $A \cup B$, is the collection of outcomes that are in either A **or** B (or both).



Example 2

- Suppose three fuses are examined by a quality control inspector to determine whether they are defective (D) or not defective (N).

  The corresponding sample space is:

  $S = \{$NNN, DNN, NDN, NND, DDN, DND, NDD, DDD$\}$.

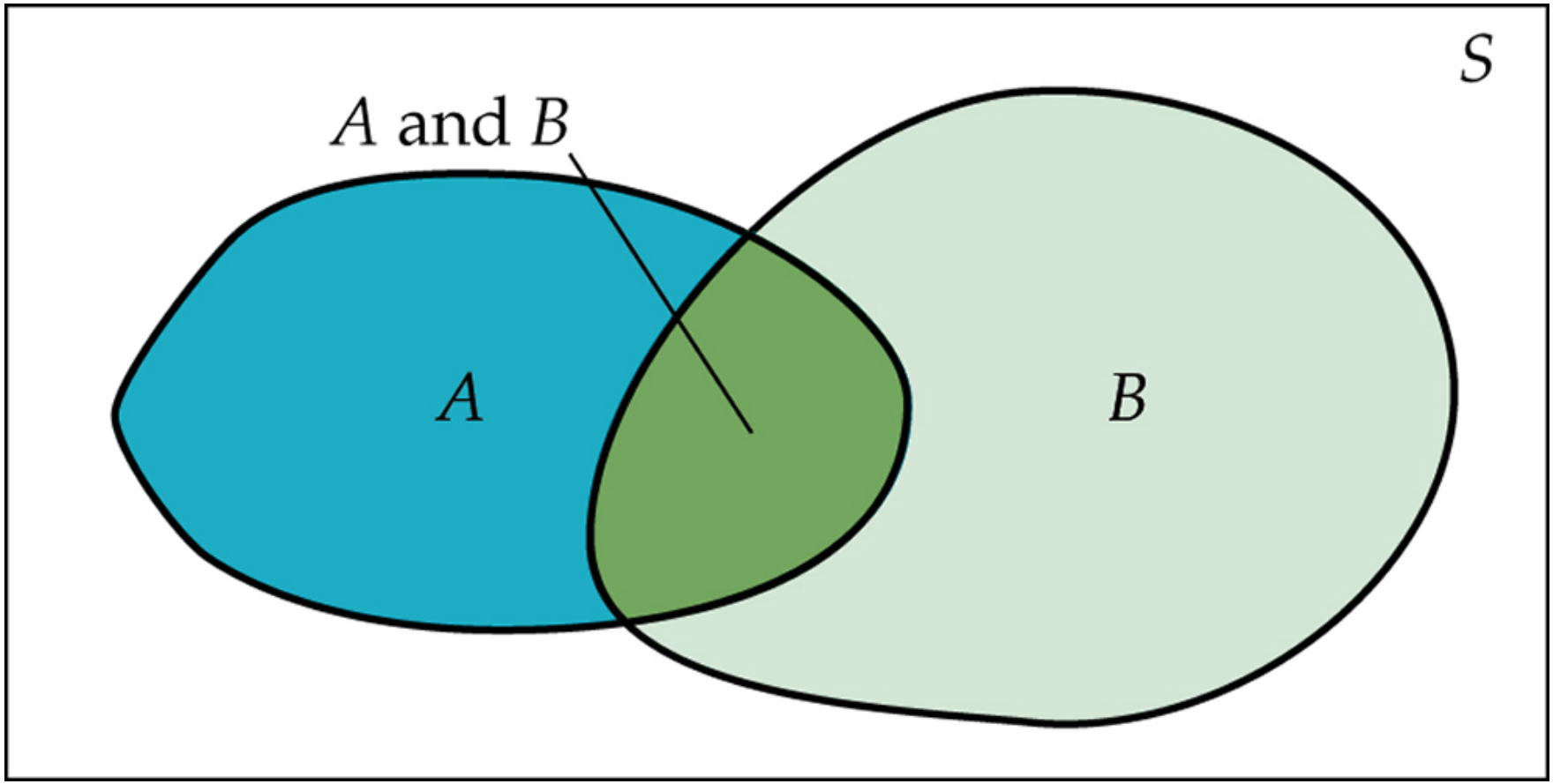
  Let E be the event that exactly two fuses are found to be defective. Let us further define the event F to be the event that exactly one fuse is found to be defective. Then what is the union of E and F?

- We have $E = \{$DDN,DND, NDD$\}$ and $F = \{$DNN,NDN,NND$\}$. So, $E \cup F$=$\{$DDN, DND, NDD, DNN, NDN, NND$\}$.

- Actually, this event represents the event where we have one or two defective fuses.

Intersection of events

- The intersection of two events A and B, denoted by $A \cap B$, consists of the outcomes that are contained in both A **and** B.



Example 3

- Suppose again two dice are rolled and we observe the number of dots facing up on each die. Define two events:

  $A =$ {at least one die shows 6 dots}
  $B =$ {the sum of all the dots is 9}

  What outcomes are in A, what outcomes are in B, what outcomes are in $A \cup B$ and what outcomes are in $A \cap B$?

- $A =$ {(1,6), (2,6), (3,6), (4,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}
  $B =$ {(3,6), (4,5), (5,4), (6,3)}.
- This gives:

  $A \cup B$={(1,6), (2,6), (3,6), (4,6), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6), (4,5), (5,4)}
  and
  $A \cap B$={(3,6), (6,3)}.

Probabilities of outcomes

- Suppose the sample space of some random experiment is $S=\{O_1, O_2, \ldots, O_n\}$, i.e., O_is denote the outcomes.

- If we denote the probability of O_i as p_i then these probabilities must satisfy:

  1. $0 \leq p_i \leq 1$ for all $i=1, 2, \ldots, n$.
  2. $p_1 + p_2 + \cdots + p_n = 1$.

- More often we use $P(O)$ to denote the probability of an outcome O (or $P(A)$ to denote the probability of some event A) and there are four properties, including these, about probability.

Properties of probability

- All probabilities must satisfy the following rules:
  1. Any probability, denoted by P, is a number between 0 and 1.
  2. All possible outcomes for some random phenomenon together must have probability 1.
  3. If two events have no outcome in common, the probability that one or the other occurs is the sum of their individual probabilities.
  4. The probability that an event does not occur is 1 minus the probability that the event does occur.

- We can write these properties in a more mathematical way:

More mathematically...

- The rules (axioms) of probability are:
  Rule 1: The probability $P(A)$ of any event A satisfies $0 \leq P(A) \leq 1$.
  Rule 2: If S is the sample space in a probability model, then $P(S)$=1.
  Rule 3: If A and B are disjoint (i.e., $A \cap B = \emptyset$) then $P(A \cup B)$=$P(A)$+$P(B)$. This is the union law for disjoint events. Another term for disjoint events is mutually exclusive events.
  Rule 4: The complement rule states that $P(A^c)$=1-$P(A)$.
- Rule 3 can be generalized: For any two events A and B

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

  This is the general addition rule (union law). The simplified version as seen above comes from the facts that when events are disjoint, i.e., $A \cap B = \emptyset$ and also, $P(\emptyset)$=0.

Equally likely model

- In general, for some random phenomenon, to find the probability of an event, you add up the probabilities of all the outcomes in the event of interest.
- The above rule can be simplified if all outcomes in the sample space are equally likely, in which case, for an event A (a subset of S), we have:

$$P(A) = \frac{\#\ \text{outcomes in}\ A}{\#\ \text{outcomes in}\ S},$$

  which is also sometimes written as

$$P(A) = \frac{N(A)}{N(S)},$$

  where $N(A)$ denotes the number of outcomes in A and $N(S)$ denotes the number of outcomes in S.
- This formula is called the equally likely or equal likelihood model.

Example 4

AN *EVENT* IS A SET OF ELEMENTARY OUTCOMES. THE PROBABILITY OF AN EVENT IS THE SUM OF THE PROBABILITIES OF THE ELEMENTARY OUTCOMES IN THE SET. FOR INSTANCE, SOME EVENTS IN THE LIFE OF A TWO-DICED ROLLER ARE:

EVENT DESCRIPTION	EVENT'S ELEMENTARY OUTCOMES	PROBABILITY
A: DICE ADD TO 3	{(1,2), (2,1)}	$P(A) = \frac{2}{36}$
B: DICE ADD TO 6	{(1,5), (2,4), (3,3), (4,2), (5,1)}	$P(B) = \frac{5}{36}$
C: WHITE DIE SHOWS 1	{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)}	$P(C) = \frac{6}{36}$
D: BLACK DIE SHOWS 1	{(1,1), (2,1), (3,1), (4,1), (5,1), (6,1)}	$P(D) = \frac{6}{36}$

(Gonnick and Smith, 1993)

Conditional Probability

- Suppose there is an experiment in which A and B are two events. Then $P(A)$ denotes the probability of event A occurring and $P(B)$ denotes the probability of event B occurring.

  These are unconditional probabilities.

- Imagine a situation where the experiment has been performed and you have been informed that event A has occurred.

- Now, one may ask what is the probability event B has also occurred? It is clear we have a condition now.

- We define the conditional probability, $P(B|A)$, as the probability that event B occurs, assuming that event A is known to occur; it is calculated as:

$$\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}}.$$

Example 5

- Suppose we roll a die and observe the number of dots facing up.
- Let A be the event that the number is odd and let B be the event that the number is at most 3.
- Then, using the equal likelihood model, $P(A)=P(1 \text{ or } 3 \text{ or } 5) = \frac{3}{6}$ and $P(B)=P(1 \text{ or } 2 \text{ or } 3) = \frac{3}{6}$.
- Now suppose a die was rolled once and we have been informed that A has occurred.
- We can ask, what is the probability of event B occurring given event A has occurred?

Independence

Two events A and B are said to be independent if $P(B|A) = P(B)$. In other words, A and B are independent if the knowledge that A has occurred does not change the probability that B will occur.

Similarly, if A and B are independent, then $P(A|B) = P(A)$.

It follows that A and B are independent **if and only if** $P(A \cap B) = P(A)P(B)$.

Permutations and Combinations

- Permutations and combinations are algebraic methods of counting and are heavily used in probability problems.
- Each separate *arrangement* of all or part of a set of items is called a permutation.
- The number of permutations is the number of different arrangements in which items can be placed.
- If the order of the items is changed, the arrangement is different, so we have a different permutation.

Combinations

- Combinations are similar to permutations, but combinations do not take order into account.
- So, AB and BA are different permutations but the same combination of letters.
- Suppose we have a total of n items and we can choose k of those items at a time, where $k \leq n$. The number of combinations of n items chosen k at a time is written as $\binom{n}{k}$ and given by

$$\binom{n}{k} = \frac{n!}{(n-k)!\,k!}. \tag{1}$$

Example 6

1. Find $\binom{3}{2}$, $\binom{5}{2}$, $\binom{10}{7}$.

2. I need to form a team of 10 from a class of 50 students. How many different teams can be formed?

Introduction to random variables

- Sample spaces are not the only way to describe random phenomena.
- In the study of random phenomena, we are most often interested in numerical outcomes, such as the number of heads observed in 3 tosses.
- For this reason we introduce some shorthand notation, i.e., let X denote the number of heads observed. If the outcome is TTH, then X=1; if the outcome is HTH, then X=2, and so on. In all, the possible values of X are 0, 1, 2 and 3. Here X is called a random variable.

Defining a random variable

- A random variable is a variable whose value is a numerical outcome of a random phenomenon.
- Random variables can be either *discrete* or *continuous*.
- A random variable is discrete if it takes on a finite or infinitely countable number of possible values (i.e. countable number of values) and is continuous if it takes on all possible values in an interval.
- We usually denote random variables by capital letters near the end of the alphabet, such as X or Y.

A simple example

- Suppose we draw a card from a deck of 52 cards four independent times (i.e., with replacement). Let us consider the random variable X=number of ♠ among these four cards. The possible values of X are 0, 1, 2, 3, and 4.

- This is clearly a discrete random variable.

- The list of all possible values of a random variable X are sometimes called its support or range, which we often denote by R_X.

An additional definition

- The probability distribution of a random variable X tells us what the possible values of X are and how probabilities are assigned to those values.

- It is clear a probability distribution of a random variable resembles a probability model for a random phenomena in that both consist of two components: one dealing with outcomes/values and the other dealing with probabilities of those outcomes/values.

A probability distribution

- The probability distribution of a *discrete* random variable X lists the values and their probabilities:

Value of X	x_1	x_2	x_3	$\cdots$	x_k
Probability	p_1	p_2	p_3	$\cdots$	p_k

- The probabilities p_i must satisfy two requirements:
  1. Every probability p_i is a number between 0 and 1.
  2. $p_1 + p_2 + \cdots + p_k = 1$.

- In the case of discrete random variables, instead of a table describing the distribution, a probability mass function may be used.

Probability mass functions (PMF)

- Let X be a discrete random variable with support $R_X = \{x_1, x_2, x_3 \ldots\}$.
- The probability mass function $p(x)$ of X gives $P(X = x_i)$ for $i = 1, 2, 3, \ldots$.
- We define the probability mass function $p(x)$ of X by

$$\boxed{p(x) = P(X = x); p(x) \geq 0 \ \forall x \in R_X}.$$

- Clearly it must be the case that $\sum_{x \in R_X} p(x) = 1$.

Cumulative distribution functions

- For a discrete random variable, the cumulative distribution function (CDF) $F(x)$ is defined to be

$$\boxed{F(x) = P(X \leq x) = \sum_{\forall y \in R_X \leq x} p(y)}.$$

- The cumulative distribution function $F(x)$ denotes the probability that the random variable X has a value less than or equal to x.

- The CDF satisfies the following properties:
  1. F is a nondecreasing function.
  2. $\lim_{x \to \infty} F(x) = 1$.
  3. $\lim_{x \to -\infty} F(x) = 0$.

Example 7

- Suppose a fair die is to be rolled and the number of dots observed, call this X.
- Then X takes on the values 1 through 6, each with probability $1/6$.

X	1	2	3	4	5	6
$p(x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

Example 7 cont'd

- Or we can write it in the form of a function:

$$p(x) = P(X = x) = \begin{cases} \frac{1}{6} & \text{if } x = 1, 2, 3, 4, 5, 6, \\ 0 & \text{otherwise.} \end{cases}$$

- Furthermore its CDF is:

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x < 1, \\ \frac{1}{6} & \text{if } 1 \leq x < 2, \\ \frac{2}{6} & \text{if } 2 \leq x < 3, \\ \frac{3}{6} & \text{if } 3 \leq x < 4, \\ \frac{4}{6} & \text{if } 4 \leq x < 5, \\ \frac{5}{6} & \text{if } 5 \leq x < 6, \\ 1 & \text{if } x \geq 6. \end{cases}$$

Continuous random variables

- If a random variable X can take on any of a continuum of values, we cannot define it by listing values x_i and giving the probability $p(x_i)$ for any single value x_i (since $P(X = x_i) = 0$).

- But we can define the cumulative distribution function

$$\boxed{F(x) = P(X \leq x)}$$

- Also, it is the case that

$$\boxed{P(a < X < b) = F(b) - F(a)}$$

and

$$\boxed{P(X > a) = 1 - F(a)}$$

Continuous random variables

- The probability density function (PDF) is defined as

$$\boxed{f(x) = F'(x) = \frac{d}{dx}F(x).}$$

- This comes from the definition of a derivative. Consider

$$\boxed{f(x)\,dx = P(X \in [x, x + dx]) = F(x + dx) - F(x)}.$$

Example 8

A continuous random variable X has the CDF

$$F(x) = \begin{cases} 0, & x \leq 0 \\ \dfrac{x^2}{9}, & 0 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

1. Find the PDF of X.
2. Find $P(X \leq 2)$.
3. Find $P(-1 < X < 2)$.
4. Find the median of X.

Example 9

A discrete random variable X has the PMF

$$p(x) = \begin{cases} c(8-x) & \text{if } x = 0, 1, 2, 3, 4, 5, \\ 0 & \text{otherwise.} \end{cases}$$

1. Find the constant c.
2. Find CDF of X.
3. Find $P(X \leq 2)$.
4. Find $P(X \geq 1)$.
5. Find $P(2 < X < 5)$.
6. Find $P(2 \leq X \leq 5)$.

Expected value

The expected value of a discrete random variable X with support $R_X = \{x_1, x_2, x_3, \ldots, x_m\}$ is defined as

$$E(X) = \sum_{i=1}^{m} x_i \, p(x_i)$$

This is also called the mean of the random variable X and is often denoted as μ or μ_X.

Find $E(X)$ in Example 9.

Expected value cont'd

- If X is a discrete random variable with PMF $p(x)$ and g is any function, then $g(X)$ is a discrete random variable and

$$\boxed{E[g(X)] = \sum_{i=1}^{m} g(x_i)\ p(x_i)}.$$

- For example, for a discrete random variable, the expected value of X^2 is

$$E(X^2) = \sum_{i=1}^{m} x_i^2 p(x_i)$$

Variance

- The variance of a random variable X is:

$$\boxed{Var(X) = E[(X-\mu)^2)] = E(X^2) - (E(X))^2}.$$

- For a discrete random variable X with support $R_X = \{x_1, \ldots, x_m\}$ this becomes:

$$\begin{aligned} Var(X) = E[(X-\mu)^2] &= \sum_{i=1}^{m} p_i (x_i - \mu)^2 \\ &= \sum_{i=1}^{m} p_i (x_i^2 - 2\mu x_i + \mu^2) \\ &= \sum_{i=1}^{m} p_i x_i^2 - 2\mu \sum_{i=1}^{m} p_i x_i + \mu^2 \\ &= E(X^2) - \mu^2 \\ &= E(X^2) - (E(X))^2. \end{aligned}$$

Find $Var(X)$ in Example 9.

Some useful properties of $E(X)$ and $Var(X)$

- Expectation is linear: $E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E(X_i)$.
- For a constant c,
  - $E(c) = c$, $E(cX) = cE(X)$ and E(c + X) = c + E(X)
  - $Var(X + c) = Var(X)$ and $Var(cX) = c^2 Var(X)$.
- If X & Y are independent, then $Var(X \pm Y) = Var(X) + Var(Y)$.

Some special probability distributions

- Some common discrete probability distributions are:
  - Bernoulli Distribution
  - Binomial Distribution
  - Poisson Distribution
- Quite often discrete random variables are counting variables.
- Some common continuous probability distributions are:
  - Normal Distribution
  - Uniform Distribution
  - Exponential Distribution

Bernoulli distribution

- A random variable X that assumes only the values of 0 (failure) or 1 (success) is known as a Bernoulli variable and the PMF is known as the Bernoulli distribution.

- It is denoted by $X \sim Bernoulli(p)$ where $P(X = 1) = p$ and $P(X = 0) = 1 - p$ for $0 \leq p \leq 1$; we can write the PMF as:

$$\boxed{P(X = x) = \begin{cases} p^x(1-p)^{1-x} & \text{if } x = 0, 1, \\ 0 & \text{otherwise.} \end{cases}}$$

- Also:

$$E(X) = p \text{ and } Var(X) = p(1-p).$$

Binomial distribution

- Suppose n independent Bernoulli trials are performed. If, for each, the probability of success is p and the probability of failure is $1-p$, then X is said to be a binomial random variable with parameters (n, p).
- It is denoted by $X \sim Bin(n, p)$ and its PMF can be written as:

$$\boxed{p(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{if } x = 0, 1, \ldots, n, \\ 0 & \text{otherwise.} \end{cases}}$$

- Also:

$$E(X) = np \text{ and } Var(X) = np(1-p).$$

- Note that if we take $n = 1$, we get the Bernoulli distribution (i.e., the Bernoulli distribution is a special case of the Binomial distribution).

Poisson distribution

- A random variable X that takes on the values 0,1,. . ., is said to be a Poisson random variable with parameter λ for some $\lambda > 0$ if its PMF is of the form:

$$\boxed{p(x) = \begin{cases} \dfrac{\lambda^x}{x!} e^{-\lambda} & \text{if } \ x = 0, 1, 2, \ldots, \\ 0 & \text{otherwise.} \end{cases}}$$

- It is denoted by $X \sim Poi(\lambda)$.
- Also we have:

$$E(X) = \lambda \ \text{ and } \ Var(X) = \lambda.$$

- The Poisson distribution is important in modelling discrete events occurring in time or in space, and it is often used when events are rare.

Poisson distribution (cont'd)

- A Poisson random variable X counts the number of occurrences of an event in a specific unit of time or area of space.
- λ is the rate at which events occur (in a single unit of time or area of space)
- We assume:
  - The probability of occurrence of the event in a specific unit of time or area of space is **constant** for any given unit of time or space.
  - The number of events that occur in any interval of time or area of space is **independent** of any other interval or area.
  - If the mean number of events for a single unit of time or space is λ, then the mean number of events which occur in t units of time or space is λt.

Normal distribution

- X is a Normal random variable with mean μ and Variance σ^2 if its PDF is given by

$$\boxed{f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}$$

  for $-\infty < x < \infty$ and $\sigma^2 > 0$.

- It is denoted by $X \sim N(\mu, \sigma^2)$.

- We also have:

$$E(X) = \mu \text{ and } Var(X) = \sigma^2.$$

Uniform distribution

- X is a Uniform random variable on the interval $[a, b]$ if its PDF is given by

$$\boxed{f(x) = \frac{1}{b-a}}$$

for $a \leq x \leq b$.

- It is denoted by $X \sim \text{Uniform}(a, b)$ or $X \sim U(a, b)$.
- We also have:

$$E(X) = \frac{a+b}{2} \quad \text{and} \quad Var(X) = \frac{(b-a)^2}{12}$$

Exponential distribution

- $X(> 0)$ is an exponential random variable with rate parameter λ, where $\lambda > 0$, if its PDF is given by

$$\boxed{f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0, \\ 0 & \text{otherwise.} \end{cases}}$$

- For its mean and variance, we have:

$$E(X) = \frac{1}{\lambda} \text{ and } Var(X) = \frac{1}{\lambda^2}.$$

- It is denoted by $X \sim Exp(\lambda)$.
- The cumulative distribution function, CDF, of X is

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}.$$

Example 10

According to the Population Survey conducted by the U.S. Census Bureau, 40% of the U.S. population 25 years old and over have completed a bachelor's degree or more. For a random sample of 50 people 25 years old or over, let X be the number of people who have completed at least a bachelor's degree.

1. What is the suitable distribution for X?
2. What is the probability of having exactly 20 people in the sample who have completed at least a bachelor's degree?
3. What is the expected number of people who have completed a bachelor's degree or more?

Example 11

Suppose, in the analysis of accidents in an industrial plant, it is known that the average number of accidents in a year is λ=1. Let X denote the number of accidents in a year.

1. What is the suitable distribution for X?
2. What is the probability that there is no accident in the next year?
3. What is the probability there are at least 3 accidents in a year?
4. What is the probability of a total of 3 accidents occurring over the next two years?

Example 12

Suppose that the length of a telephone call (in minutes) is exponentially distributed with rate parameter $\lambda = 0.2$.

1. What is the probability that the call lasts between 2 and 6 minutes?
2. Find the mean, median, the first and third quartiles, and the interquartile range of the call length.

Example 13 (Review)

(a) Suppose that a random variable X is normally distributed with mean 50 and standard deviation 3. What is $P(48 < X < 53)$?

(b) Suppose that vault scores at a gymnastics competition are normally distributed with mean 8.2 and standard deviation 0.4. The top 6% of gymnasts go to nationals. What is the minimum vault score required to go to nationals?

Statistical distributions and related functions in R

- For common distributions, R has built-in functions:

Binomial	pbinom	qbinom	dbinom	rbinom
Poisson	ppois	qpois	dpois	rpois
Normal	pnorm	qnorm	dnorm	rnorm
Uniform	punif	qunif	dunif	runif
Exponential	pexp	qexp	dexp	rexp

- In these functions, there is a root name, and the first letter prefix stands for:

  p for "probability", the CDF

  q for "quantile", the inverse CDF (quantile function)

  d for "density", the density (PDF or PMF)

  r for "random", a random value from the distribution

- For details on any of these, including the necessary arguments, see their help files.

Sample code

- We can use these functions for a variety of purposes, such as finding means or medians, finding probabilities or generating samples.
- Consider finding the median of an exponential distribution with rate parameter λ=2:

```
> med.exp2 <- qexp(0.5,2)
> med.exp2
[1] 0.3465736
```

- Consider a binomial random variable with parameters n=10 and p=0.8. Then, $P(X \leq 3)$ and $P(X \leq 8)$ can be found using:

```
> p.x3 <- pbinom(3,10,0.8)
> p.x3
[1] 0.0008643584
> p.x8 <- pbinom(8,10,0.8)
> p.x8
[1] 0.6241904
```

Sampling from standard statistical distributions

- Suppose we wish to draw a sample from a population (probability distribution); this can be thought of as data generation.
- You can draw a sample from a distribution to get an idea of the shape of a distribution.
- Suppose we wish to see the effect of increasing sample sizes on the shape of Normal(4,36) data; we can use R to do this:

```
> norm.data1 <- rnorm(25,4,6)
> norm.data2 <- rnorm(50,4,6)
> norm.data3 <- rnorm(100,4,6)
> norm.data4 <- rnorm(500,4,6)
```

- Making histograms of each:

```
par(mfrow=c(2,2))
hist(norm.data1,main="n=25")
hist(norm.data2,main="n=50")
hist(norm.data3,main="n=100")
hist(norm.data4,main="n=500")
dev.off()
```

Sampling from non-standard statistical distributions

- For simple/standard distributions, it is easy to obtain samples from it as there are built-in routines for simulating from them.
- For some distributions, there may not be built-in routines. Then what do we do?
- Other ways to generate samples from any probability distribution include the R function sample and the inverse CDF method, also known as the inversion method.

sample

- Suppose that we want to draw a sample from some discrete probability distribution.
- One option in R is to use the function sample.
- According to its help file, it has 4 important arguments:

```
sample(x, size, replace = FALSE, prob = NULL)
```

`x`	either a vector of one or more elements from which to choose, or a positive integer. See 'Details.'
`size`	a non-negative integer giving the number of items to choose.
`replace`	should sampling be with replacement?
`prob`	a vector of probability weights for obtaining the elements of the vector being sampled.

Example 14

- Suppose that we wish to draw from the distribution from the dice example on Slide 35 (which is equivalent to rolling a die!).

- Suppose we want to sample 5 values (roll a die 5 times); then, using sample:

```
> dice.data <- sample(x=1:6,size=5,replace=TRUE,prob=rep(1/6,6))
> dice.data
[1] 6 4 2 6 5
```

- To use this method (and function), one needs to be able to supply the support (values to choose from) and their probabilities.

Inversion method: continuous distributions

- Suppose that we want to draw a sample from a probability distribution with PDF $f(\cdot)$ and corresponding CDF $F(\cdot)$.
- The algorithm for the inversion method involves the following steps:
  1. Draw a random number from $U \sim U(0, 1)$.
  2. Then $X = F^{-1}(U)$ is a draw from $f(\cdot)$.
- To use this method, one needs to be able to invert the CDF.

Example 15

- Suppose that we want to sample from $exp(\lambda)$ using the inversion method.

- First, we note $F(x) = 1 - e^{-\lambda x}$. Set $u = 1 - e^{-\lambda x}$

$$\Rightarrow x = F^{-1}(u) = -\frac{ln(1-u)}{\lambda}.$$

- Then:
  1. Draw a random number from $U \sim U(0, 1)$.
  2. $x = -\frac{ln(1-u)}{\lambda}$ is an observation from $exp(\lambda)$.

The inversion method: discrete distributions

- Assume that X is a discrete random variable with support $R_X = \{x_1, \ldots, x_m\}$ where $x_1 < x_2 \cdots < x_m$ and such that $P(X = x_k) = p_k$, $k = 1, \ldots, m$.

- The algorithm is as follows:

  1. Draw a random number from $U \sim U(0, 1)$.

  2. Determine if $U \leq p_1$.
     - If $U \leq p_1$, then return $X = X_1$.

     - Otherwise, determine the index k such that $\sum_{j=1}^{k-1} p_j < U \leq \sum_{j=1}^{k} p_j$ and return $X = x_k$.